IS101 — Introduction to Computing

SOC 101 Grade Book

College of Southern Nevada · Unit 3 — Spreadsheets · Variation 1 · 100 points · 26 tasks

How to use this document: Download the starter workbook (Excel_Var1_Starter.xlsx) from the Project Grader page. Open it in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order — the tasks are sequenced so every formula has the data it needs. The Grade Scale on the second sheet tab uses point values based on 300 total points possible: 270+ = A, 240+ = B, 210+ = C, 180+ = D, below 180 = F. When you're done, save the file and upload it to the Variation 1 grader for instant scoring. **Important:** When writing formulas, always reference cell addresses (like H4), not column names or words. Typing a word like “Average” or “Total” in a formula causes Excel to auto-convert it into an error.

Setup Tasks 1–5 · 19 points

Task 1 3 points

Freeze the header row and the student-name column

WHY THIS MATTERS

Freezing keeps headers and student names visible while scrolling through scores.

WHERE TO FIND IT

Click cell B4 — this is the cell below the header row and to the right of the name columns. Then View tab → Freeze Panes → Freeze Panes. This freezes both the headers AND the name column in one step. Do NOT use “Freeze Top Row” or “Freeze First Column” — those only freeze one or the other.

HOW YOU'LL KNOW IT WORKED

When you scroll down, the header row stays visible. When you scroll right, the name columns stay visible.

Task 2 3 points

Add a comment on the Feedback column

WHY THIS MATTERS

Comments let you document notes without changing the data.

WHERE TO FIND IT

Right-click any cell in the Feedback column (column M) → Insert Comment (or New Note) → type your note.

HOW YOU'LL KNOW IT WORKED

A red triangle appears in the cell corner; hovering shows your comment.

Task 3 3 points

Bold and shade the header row

WHY THIS MATTERS

Bold text and a colored background make headers stand out from the data.

WHERE TO FIND IT

Select the header row (row 3) → Home tab → Bold (Ctrl+B) → Fill Color → choose a color.

HOW YOU'LL KNOW IT WORKED

The header row has bold text on a colored background.

Task 4 8 points

Center the score columns

WHY THIS MATTERS

Centering aligns numeric scores in the middle of each cell for a clean look.

WHERE TO FIND IT

Select all score columns (columns C through G) → Home tab → Alignment group → Center.

HOW YOU'LL KNOW IT WORKED

All score values are centered in their cells.

Task 5 2 points

Wrap text in the Feedback column

WHY THIS MATTERS

Wrap Text shows long feedback on multiple lines within the cell.

WHERE TO FIND IT

Select the Feedback column (column M) → Home tab → Alignment group → Wrap Text.

HOW YOU'LL KNOW IT WORKED

Long feedback text wraps to multiple lines within each cell.

Formulas & Functions Tasks 6–16 · 49 points

Task 6 6 points

Use SUM and AVERAGE to calculate each student's Total and Average

WHY THIS MATTERS

SUM adds all scores; AVERAGE calculates the mean. The Total column is H; the Average column is I. Scores are in columns C through G (HW1, HW2, Quiz, Midterm, Final). **Important:** Reference only columns C through G in both formulas. Do NOT include column H in the AVERAGE range.

WHERE TO FIND IT

Click cell H4 → type a SUM formula referencing the score cells in columns C through G for that student → Enter. Click cell I4 → type an AVERAGE formula referencing the same columns C through G → Enter. Copy both formulas down for all students (rows 4–11).

HOW YOU'LL KNOW IT WORKED

Column H shows each student's total points (e.g., 236 for the first student); column I shows their average (e.g., 47.2).

Task 7 4 points

Use COUNTA to count submitted assignments per student

WHY THIS MATTERS

COUNTA counts non-empty cells — useful for tracking how many assignments each student turned in. Place the formula in column N.

WHERE TO FIND IT

Click cell N4 → type a COUNTA formula referencing the score cells in columns C through G for that student → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Column N shows a count (like 5) for each student who submitted all assignments.

Task 8 4 points

Use an absolute reference ($) in the Percent formula

WHY THIS MATTERS

Absolute references lock a cell so it doesn't shift when you copy. The Percent column (J) divides each student's Total by the Total Points Possible value in cell P2. Use $ signs on the P2 reference so it stays locked when copied down.

WHERE TO FIND IT

Click cell J4 → type a formula that divides the student's Total in column H by the Total Points Possible cell, using $ signs on the reference to P2 → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

The formula bar shows $ signs in the denominator reference. Copying down keeps the same denominator cell for every row.

Task 9 3 points

Name the grade-scale lookup range “GradeScale”

WHY THIS MATTERS

Named ranges make formulas easier to read. The GradeScale table is on the second sheet tab and contains point cutoffs and letter grades.

WHERE TO FIND IT

Click the GradeScale sheet tab → select the range A4:B8 (the point values and letter grades, not the headers) → click in the Name Box (left of the formula bar) → type GradeScale → press Enter.

HOW YOU'LL KNOW IT WORKED

The Name Box shows “GradeScale” when the range is selected. Formulas {ARR} Name Manager lists it.

Task 10 5 points

Use IF to mark each student Pass or Fail

WHY THIS MATTERS

IF tests a condition and returns one value if true, another if false. A student passes if their Total (column H) is 180 or above (60% of 300 points). Reference a cell address (like H4), not the word “Total.”

WHERE TO FIND IT

Click cell K4 → type an IF formula that checks whether the student’s Total in column H is greater than or equal to 180. If yes, show “Pass”; if no, show “Fail.” → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Column K shows “Pass” for students with 180+ total points and “Fail” for those below.

Task 11 6 points

Use a nested IF to assign letter grades A–F

WHY THIS MATTERS

A nested IF tests multiple conditions in sequence. Use each student’s Total (column H) and the point cutoffs from the Grade Scale: 270+ = A, 240+ = B, 210+ = C, 180+ = D, below 180 = F. Start with the highest cutoff and work down. Reference cell addresses, not words.

WHERE TO FIND IT

Click cell L4 → type a nested IF formula that checks the student’s Total in column H against each cutoff, starting with 270 and working down to 180 → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Column L shows a letter grade (A, B, C, D, or F) for each student based on their total points.

Task 12 6 points

Use VLOOKUP to look up a student’s letter grade by name

WHY THIS MATTERS

VLOOKUP can search for a value in the first column of a range and return a value from another column. The “Look Up Student:” label is in cell P8. You will type a student’s last name and write a formula to retrieve their letter grade.

WHERE TO FIND IT

Type Cruz in cell P9. Click cell P10 → type a VLOOKUP formula that searches for the name in P9 within the student data range (A4:L11), returns the value from column 12 (the Letter grade column), and uses FALSE for an exact match → Enter.

HOW YOU'LL KNOW IT WORKED

Cell P10 shows “B” — the letter grade for Cruz, who has 254 total points.

Task 13 5 points

Wrap the VLOOKUP in IFERROR to handle errors

WHY THIS MATTERS

IFERROR takes two parameters: the formula to try, and what to show if it returns an error. If someone types a name that doesn't exist, the VLOOKUP returns #N/A. IFERROR catches that and shows a friendly message instead.

WHERE TO FIND IT

Edit the formula in P10 → wrap the entire VLOOKUP inside IFERROR. The first parameter is your VLOOKUP formula; the second parameter is a fallback message like "Not Found" → Enter.

HOW YOU'LL KNOW IT WORKED

P10 still shows "B" for Cruz. If you change P9 to a name that doesn't exist, P10 shows "Not Found" instead of #N/A.

Task 14 3 points

GradeScale named range

WHY THIS MATTERS

This task verifies that the GradeScale named range you created in Task 9 is present in the workbook.

WHERE TO FIND IT

If you completed Task 9, this task is already done. No additional work needed.

HOW YOU'LL KNOW IT WORKED

Formulas {ARR} Name Manager shows GradeScale in the list.

Task 15 4 points

Use COUNTIF to count how many students earned an A

WHY THIS MATTERS

COUNTIF counts cells matching a condition — useful for class statistics. Place the formula in the summary area near cell P6. Count the Letter grades in column L.

WHERE TO FIND IT

Click cell P6 (or a nearby empty cell in the summary area) → type a COUNTIF formula that counts how many cells in the Letter column (L4:L11) contain "A" → Enter.

HOW YOU'LL KNOW IT WORKED

The cell shows the count of students who earned an A. With the current data, this value should be 0 — no student has 270+ points.

Task 16 3 points

Use TODAY() to stamp the report-generated date

WHY THIS MATTERS

TODAY() inserts the current date and updates automatically each time the file opens. The "Report Generated:" label is in cell Q1.

WHERE TO FIND IT

Click cell R1 (next to "Report Generated:") → type =TODAY() → Enter.

HOW YOU'LL KNOW IT WORKED

Cell R1 shows today's date and updates whenever you open the file.

Formatting Tasks 17–21 · 14 points

Task 17 2 points

Format the Percent column with the Percent number format

WHY THIS MATTERS

Percent format displays decimal values as percentages (0.85 becomes 85%).

WHERE TO FIND IT

Select the Percent column data (column J) → Home tab → Number group → Percent Style.

HOW YOU'LL KNOW IT WORKED

Values display as 79% instead of 0.786.

Task 18 3 points

Highlight failing totals with conditional formatting

WHY THIS MATTERS

Conditional formatting visually flags students who need attention. Highlight Total scores (column H) below 180 (the minimum passing score out of 300 points).

WHERE TO FIND IT

Select the Total column data (H4:H11) → Home tab → Conditional Formatting → Highlight Cell Rules → Less Than → enter 180 → choose a red fill → OK.

HOW YOU'LL KNOW IT WORKED

Cells with totals below 180 are highlighted in red.

Task 19 3 points

Convert the grade data to an Excel Table

WHY THIS MATTERS

Excel Tables add automatic filtering, banded rows, and structured references.

WHERE TO FIND IT

Click anywhere in the grade data → use any method: Ctrl+T, Insert tab → Table, or Home tab → Format as Table → verify the range → OK.

HOW YOU'LL KNOW IT WORKED

Filter arrows appear on each column header; the Table Design tab appears when you click in the table.

Task 20 3 points

Sort by Total (descending) then Last Name (ascending)

WHY THIS MATTERS

Multi-level sorting puts the highest scores first and alphabetizes ties. “Total” means column H (the overall total), not the “Final” exam column.

WHERE TO FIND IT

Data tab → Sort → Sort by: Total (column H) → Largest to Smallest → Add Level → Then by: Last Name (column A) → A to Z → OK.

HOW YOU'LL KNOW IT WORKED

Students with the highest totals appear first; tied students are alphabetical.

Task 21 3 points

Filter to students currently failing

WHY THIS MATTERS

Filtering shows only students who need intervention. Filter the Letter column (L) to show only F grades.

WHERE TO FIND IT

Click the dropdown arrow on the Letter column header (column L) → uncheck Select All → check only “F” → OK.

HOW YOU'LL KNOW IT WORKED

Only students with an F grade are visible.

Charts Tasks 22–24 · 11 points

Task 22 4 points

Create a Column chart of class average by assignment

WHY THIS MATTERS

Charts visualize how the class performed on each assignment.

WHERE TO FIND IT

Select the assignment headers (C3:G3) and the corresponding class averages → Insert tab → Charts group → Column → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears showing average scores for each assignment.

Task 23 4 points

Add data labels and a chart title

WHY THIS MATTERS

Data labels show exact values on each bar; a chart title identifies the chart.

WHERE TO FIND IT

Click the chart → Chart Design tab → Add Chart Element → Data Labels → Outside End. Then Add Chart Element → Chart Title → type a descriptive title.

HOW YOU'LL KNOW IT WORKED

Each bar shows its value; the chart has a title at the top.

Task 24 3 points

Add alt text to the chart

WHY THIS MATTERS

Alt text describes the chart for screen readers and accessibility.

WHERE TO FIND IT

Right-click the chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED

Right-clicking the chart and selecting Edit Alt Text shows your description.

Finishing Touches Tasks 25–26 · 7 points

Task 25 4 points

Add a header with the course name and a footer with page number

WHY THIS MATTERS

Headers and footers identify printed pages. **Important:** Use the Page Setup method below, not Insert {ARR} Header & Footer, which switches to Page Layout view and can unfreeze your panes from Task 1.

WHERE TO FIND IT

Page Layout tab → click the Page Setup dialog launcher (small arrow in the bottom-right corner of the Page Setup group) → Header/Footer tab → Custom Header: type “SOC 101 Grade Book” → Custom Footer: insert page number → OK.

HOW YOU'LL KNOW IT WORKED

Print Preview shows the course name at the top and page number at the bottom. Your frozen panes from Task 1 remain intact.

Task 26 3 points

Set Title and Author workbook properties

WHY THIS MATTERS

Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT

File tab → Info → Properties panel. Set Title to “SOC 101 Grade Book” and Author to your name.

HOW YOU'LL KNOW IT WORKED

File {ARR} Info shows Title and Author populated.

Total: 100 points. Every task is graded automatically when you upload your file. © Dr. Mark Taormino · All Rights Reserved · College of Southern Nevada